Angelos Ioannis Katsampekis

Unity developer and research engineer

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PROFILE

Unity Developer and XR Research Engineer with 5+ years of experience creating immersive XR/AR/VR experiences and games. Delivered large-scale projects for **Samsung, Coca-Cola**, and **EU Horizon** programs, ranging from VR avatar-tracking systems and AI-powered museums to a physics-based drone and missile simulator with machine-learning navigation. Skilled in **Unity, C#, C++, and Python**, with practical expertise in gameplay programming, physics simulation, multiplayer networking, deep learning, generative AI, and performance optimization.

EDUCATION

2020 — **Physics (BSc)**
Aristotle University of Thessaloniki

2025 — **Advanced Computer and Communication Systems (MSc)**
Aristotle University of Thessaloniki

2021 — **Data Science with Python (Certificate)**
National and Kapodistrian University of Athens

PROFESSIONAL EXPERIENCE

11/2022 – 08/2025
Thessaloniki, Greece
Unity Developer – Research Assistant
CERTH – The Centre for Research & Technology, Hellas

01/2025 – 08/2025
Thessaloniki, Greece
Software Developer
Pragma – IoT Solutions

PROJECTS

Drone & Missile Simulator (Unity)

- Designed and built a complete **physics-based flight simulator** end-to-end, from core architecture and aerodynamics modeling to UX and control systems.
- Developed and optimized a custom **multithreaded aerodynamic solver** with per-vertex force computation, parallelized using Unity Jobs and Burst-compiled C#, achieving ~6× faster performance (2 ms vs 13 ms per frame on i7-13700K) compared to single-threaded simulation for precise real-time flight dynamics.
- Trained **autonomous drones** via physics-based deep **reinforcement learning** (Unity ML-Agents) with depth-camera vision input, performing over 50 million training steps to reach 95% navigation accuracy in dense forest environments.
- Generated missile telemetry datasets for AI targeting and machine-learning research.

Samsung Electronics – 3D Avatar Body Capture (Unity)
CERTH - Samsung Electronics

- Prototyped an **AI-driven VR full-body avatar estimation system** trained on motion-capture data (AMASS), predicting full-body animation in real time from only head and hand tracking inputs.
- Integrated and benchmarked multiple inference engines (Sentis, ONNX Runtime, TensorFlow Lite) across mobile XR devices, achieving up to 2× improvement in inference performance and **sub-1 ms latency per prediction**, enabling smooth real-time avatar motion on standalone headsets.
- Optimized neural network deployment for cross-platform XR, balancing accuracy, latency, and efficiency under mobile GPU/CPU constraints.

Samsung Electronics – Mobile Game Optimization
Pragma IoT Solutions - Samsung Electronics

- Developed Python and C++ tools for Samsung that convert and package ONNX models to Android Neural Networks (NNAPI) for mobile AI inference, streamlining model deployment on Android devices.
- Benchmarked **mobile games** using a custom graphics monitoring harness, capturing frame-time, thermal, and battery data to guide **performance optimization** in collaboration with Samsung's engineering team.

5G VR Automotive Tour (Unity Multiplayer)
CERTH

- Developed a **multiplayer VR** application in Unity that receives a live 360° video stream from a real vehicle over 5G and projects it to VR users in real time.
- Implemented **foveated rendering** and AI-based face anonymization, maintaining visual fidelity while reducing bandwidth usage by up to 40%.
- Collected and analyzed over 10,000+ network performance samples (latency, jitter, throughput) to evaluate 5G streaming stability for immersive applications.

Palimpsisto – 3D Geospatial Platform (Unity)
CERTH

- Developed an interactive geospatial platform for the archaeological site of Kythnos, funded by the Greek government.
- Integrated Cesium to stream **photogrammetry scans** as 3D tiles from a geospatial database through a **REST API** backend using **JSON** data serialization.
- Implemented an AI assistant for in-platform interaction.
- Integrated live museum metadata through a REST API, enabling curators to update and manage exhibits dynamically.

ReEvaluate – AI-Driven VR Museum (EU Horizon, Unity)
CERTH

- Developed an immersive VR museum where **generative AI** produces themed textures and automatically places artifacts based on semantic descriptions and REST API metadata integration.

VirTourArt Platform (EU Horizon, Unity SDK)
CERTH

- Developed and delivered AR/VR rendering and multimodal interaction modules with localization and 3D reconstruction capabilities, integrating them into a cross-platform Unity SDK for **XR applications**.

EMBNOESIS – Intelligent NB-IoT sensors for the construction industry
CERTH

- Developed a Python-based **data visualization tool** that converts live NB-IoT sensor streams into interactive dashboards for construction site monitoring.

Coca-Cola CCH: Software Development Service
CERTH - Coca-Cola

- Designed and implemented **machine learning algorithms** for Coca-Cola, collaborating with their data science team to support inventory optimization research.

SKILLS

Unity Engine
C#, SRP, URP, HDRP, XR Interaction Toolkit, DOTS, Burst, Jobs

VR/AR Development
Meta Quest, Valve Index, OpenXR, Meta XR SDKs

Gameplay & Systems Programming
Physics Simulation, Multithreading, Optimization

Multiplayer Networking
Netcode for GameObjects, 5G Streaming

Machine Learning Integration
ML-Agents, Deep Reinforcement Learning, Generative AI Pipelines

Mobile Optimization
GLES, Vulkan, Android NDK, Profiling & Performance Benchmarking

Native & Tooling Development
C++, Python Tooling, REST API Integration, Git, CI/CD Automation

Model Deployment
ONNX, NNAPI, TensorFlow Lite, Sentis

Geospatial Visualization
Cesium 3D Tiles, Geospatial Data, REST APIs

Data Visualization
NB-IoT Dashboards, Analytics, Python Visualization Tools

LANGUAGES

- Greek
- English